No-Cloning Theorem: Key Gotchas and Intuition

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Purpose

This document summarizes the essential conceptual points, pitfalls, and "gotchas" related to the **no-cloning theorem** in quantum mechanics, useful for exams and conceptual understanding.

Basic Statement

- The **no-cloning theorem** states:

  There is no universal unitary operator U such that $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$.

- Only **orthogonal states** can be perfectly cloned.
- Approximate or probabilistic cloning is possible, but perfect universal cloning of unknown quantum states is forbidden.

Gotchas and Common Intuition Traps

- **Classical vs Quantum**: Classical information can be copied freely, quantum states cannot.
- **Known states**: If you know the state exactly, you can prepare a copy. No-cloning restricts *unknown states*.
- **Entanglement preservation**: Cloning a part of an entangled state generally destroys entanglement correlations.
- **Linearity of quantum mechanics**: No-cloning is a direct consequence of linearity:

$$U(\alpha|0\rangle + \beta|1\rangle)|0\rangle \neq \alpha U|0\rangle|0\rangle + \beta U|1\rangle|0\rangle = (\alpha|0\rangle + \beta|1\rangle)^2$$

- **No-deleting theorem**: Complementary to no-cloning — unknown quantum states cannot be deleted by a unitary operation.
- **Perfect cloning implies superluminal signaling**: If cloning were allowed, one could violate causality via entangled states.
- **Approximate cloning**: There exist optimal approximate cloners, but fidelity is always less than 1.

- **State-dependent cloning**: Cloning is allowed if the set of states is known and linearly independent.
- **No universal broadcasting for mixed states**: Only commuting density matrices can be perfectly broadcast.
- **Cloning vs measurement**: Measuring a quantum state destroys the original state; cloning does not bypass the measurement disturbance.

One-Line Memory Triggers

- "No universal cloning of unknown states."
- "Orthogonal states can be cloned."
- "Approximate cloning is possible, perfect cloning is not."
- "No-cloning arises from linearity of quantum mechanics."
- "Cloning unknown states $\Rightarrow$ superluminal signaling."
- "State-dependent cloning allowed only for known, linearly independent states."
- "No universal broadcasting for non-commuting mixed states."
- "No-deleting theorem complements no-cloning."